

Coral-Associated Fishes Accelerate Coral Wound Healing and Photosynthetic Recovery

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ABSTRACT

Coral resilience is increasingly recognized as the product of complex interactions between corals, their endosymbiotic algae, and external symbionts such as fishes and invertebrates. While previous studies have documented how coral-associated fishes influence coral growth and thermal tolerance, their role in supporting tissue repair and regeneration following physical injury remains poorly understood. Here, we experimentally test whether coral-associated fishes facilitate wound healing in the coral *Pocillopora* spp. We imposed small and large wounds and monitored healing in the presence and absence of resident damselfishes. Corals exposed to fishes exhibited faster wound closure and higher photosynthetic efficiency in recently regenerated tissue, particularly for larger wounds. These findings suggest that coral-associated fishes promote the energetic and microenvironmental conditions necessary for tissue repair. Our results extend the emerging understanding of coral holobiont resilience, highlighting the importance of external symbionts in sustaining coral recovery from physical disturbances.

Keywords: Coral regeneration, Mutualism, Damselfish, *Pocillopora*, Wound healing, Coral resilience, *Dascyllus flavicaudus*

INTRODUCTION

Coral reefs are increasingly understood as complex symbiotic systems not only defined by the well-known partnership between corals and their algal symbionts, but also shaped by a broader network of external associations with fishes and invertebrates. Many of these coral-associated fishes and invertebrates live in an intimate relationship with corals, influencing the physical and chemical environment directly surrounding coral tissues⁽¹⁾. These external partners have been shown to affect nutrient cycling⁽²⁾, predator-prey dynamics⁽³⁾, sediment removal^(4,5), parasite loads⁽⁶⁾, and water flow⁽⁷⁾, collectively affecting coral physiology, growth, and survival^(2,8,9). As research on these external symbioses grows, it is becoming clear that the resilience of reef-building corals may depend as much on these ecological partnerships as on their internal symbioses^(as reviewed in 10).

Despite this growing appreciation, much of the focus to date has centered on how these associations influence coral responses to large-scale, chronic stressors such as thermal bleaching and ocean acidification^(2,11,12). However, corals also routinely face localized, physical injury from predation, storms, disease, and human activity⁽¹³⁾. The ability to heal tissue following injury is a fundamental component of coral resilience: e.g., if left unhealed, a wound can be colonized by macroalgae or pathogens, potentially threatening the survival of the entire colony⁽¹⁴⁻¹⁶⁾. However, wound healing and tissue regeneration can impose substantial energetic costs that compete with other metabolic demands: damaged corals exhibit reduced growth rates, compromised reproductive output, and depleted lipid reserves, particularly in branching species like *Pocillopora*^(17,18). While we know that intrinsic factors such as wound size, energetic state, and algal symbiont density influence healing rates⁽¹³⁾, far less is known about how the surrounding biotic environment may shape these repair processes. In particular, the potential role of coral-associated fishes to facilitate or constrain wound healing remains virtually unexplored^(but see 19 for a notable exception).

Multiple mechanisms suggest that coral-associated fishes may contribute to wound repair. By excreting ammonium-rich waste, these fishes provide nitrogen that can be readily assimilated by algal symbionts^(2,9), potentially enhancing photosynthetic productivity and fueling the energetically demanding process of tissue regeneration. Their territorial behaviors also reduce exposure to corallivores and other physical disturbances that could disrupt wound closure^(20,21). Furthermore, their constant movement within coral branches enhances localized water flow and oxygenation, especially at night when respiration can drive hypoxia inside branching coral colonies^(7, 22,23). This additional oxygen delivery could help sustain the high metabolic demands of wound healing under conditions that would otherwise limit tissue repair. These

distinct pathways suggest that coral-associated fishes could modify the physical and energetic microenvironment in ways that promote regeneration, yet their role in this process has not been empirically tested.

Here, we directly test whether coral-associated fishes facilitate wound healing in their hosts. We hypothesized that coral-associated damselfish enhance both the rate of wound healing and the physiological recovery of host corals. Using colonies of *Pocillopora* spp., we inflicted small and large wounds and monitored healing trajectories in the presence and absence of resident damselfishes. This experiment provides a novel test of how external symbiotic partners contribute to coral regeneration, and more broadly, illuminates the functional importance of coral-associated biodiversity in promoting resilience to physical disturbance.

METHODS

Study Site and System

We conducted this experiment in August 2024 at the Richard B. Gump South Pacific Research Station (17°30'S, 149°50'W) on the northern shore of Mo'orea, French Polynesia. The site features a mosaic of shallow patch reefs consisting of, among others, the branching coral *Pocillopora*. Species-level identification of *Pocillopora* could not be visually or genetically resolved, and is so referred to by genus. Patch reefs are interspersed with sand and coral rubble. *Pocillopora* colonies provide structural habitat for a range of coral-associated species, including trapeziid crabs and damselfishes, which can influence coral physiology and resilience^(1,2,12,21). Among this community of exosymbionts is the yellowtail damselfish (*Dascyllus flavicaudus*), which is one of the most abundant fishes that occupy *Pocillopora*. We therefore quantified the effect of *D. flavicaudus* on wound regeneration and photosynthetic efficiency in *Pocillopora* spp.

Specimen Collection and Experimental Setup

We collected 72 coral fragments from mature *Pocillopora* spp. colonies (>20 cm diameter) without damselfish from a reef in the lagoon on the north shore. From the same reef, we also collected 18 adult *Dascyllus flavicaudus*. We transported corals and fish separately in seawater-filled coolers to the laboratory, where we mounted each coral fragment on a ceramic frag plug (Oceans Wonders) using marine epoxy (Z-Spar, Kopcoat, Pittsburgh, PA, USA). We randomly distributed 12 fragments across each of six 95 L flow-through aquaria supplied with unfiltered seawater at a rate of ~6 L h⁻¹. We held fish separate from corals during the acclimation period.

Following a 3-day acclimation under ambient light and temperature conditions, we assigned 4 fragments in each tank to one of three treatments: control (no wound), small wound (avg: 1.48 ± 0.398 cm²), or large wound (avg: 3.82 ± 1.08 cm²). We created wounds on the upper surface of fragments using an airbrush and a circular stencil to remove surface tissue while standardizing wound size and location. Immediately after wounding, we randomly introduced six *D. flavicaudus* (total biomass ~36 g) into half of the tanks (n = 3). Tanks without fish served as controls.

We fed all tanks—regardless of fish presence—2 g of TetraMarine Saltwater Flakes (Tetra US, Blacksburg, VA, USA) three times per day (morning, midday, and evening). We removed uneaten food after 10 minutes via a siphon. We scrubbed tanks and coral plugs daily to prevent algal overgrowth.

Monitoring Wound Healing, Growth, and Photosynthetic Performance

To assess wound healing and tissue regeneration, we photographed all corals with an Olympus TG-6 digital camera and ruler scale bar immediately post wounding. We then randomly selected one fragment from each wounding treatment per tank for daily photography (Figure 1). On day 21, we photographed all wounds again. Using the initial images, we measured the initial wound size (cm²) using ImageJ. Using the final images, we quantified the wound area that remained unhealed, which we determined by the absence of visible tissue, regardless of polyp formation. We then calculated the healing rate (cm²/day) as the change in wound size (initial-final) divided by the duration of the experiment: 21 days.

To assess coral growth, we measured the buoyant weight of each coral at the start (day 0) and end (day 21) of the experiment⁽²⁴⁾. At the end of the experiment we also measured the surface area of each fragment by wax dipping⁽²⁵⁾. For each wounded fragment, initial wound size was subtracted from total surface area measurements to provide an accurate measure of the initial surface area of live coral. We calculated coral growth (mg/cm²/day) as the area-adjusted calcification rate: (final buoyant weight - initial buoyant weight)/((surface area - initial wound area) * 21).

To assess coral physiological status on day 21, we measured dark-adapted photosynthetic efficiency (Fv/Fm) using pulse-amplitude modulated fluorometry (Diving-PAM, Walz, Germany)⁽²⁶⁾. For corals without wounds, measurements were taken near the upper face of the fragment in the same region where we made wounds in the other corals; for wounded corals, measurements were taken near the center of the wound. Throughout the healing process, turf algae colonized open wounds. By day 21, algae persisted on the wounds of eight corals. We therefore carefully removed the algae with tweezers before taking the PAM measurements. Of these eight corals, seven were in tanks without fish and six had large wounds. We compared the corals with algae to those without algae, and since Fv/Fm did not differ significantly between these groups, (p = 0.106), the results we present below do not distinguish between the corals that were vs. were not colonized by algae.

Statistical Analyses

We performed all statistical analyses in R (v4.3.1). To assess wound healing rate (cm²/day), we fit linear mixed-effects models—with wound healing rate as the response variable—using the lme4 package. Fixed effects included fish presence (present or absent), wound size (small, large), and their interaction, with tank ID included as a random effect to account for non-independence.

To analyze fish and wound effects on coral skeletal growth, we used an ANCOVA-style approach where we included log(Initial Mass) as a covariate in a full model with treatment effects. This approach simultaneously estimates the allometric coefficient (*b*) while testing for treatment effects, avoiding the bias that would occur if *b* were estimated from pooled data ignoring treatments. We fit a linear mixed-effects model with log(Final Mass) as the response variable: $\log(\text{Final Mass}) = \alpha + b \times \log(\text{Initial Mass}) + \text{Fish} + \text{Wound} + \text{Fish} \times \text{Wound} + \text{Tank}$ where Fish (present or absent) and Wound (none, small, large) were fixed effects, and Tank was included as a random intercept to account for non-independence of corals within the same tank. We then used maximum likelihood estimation and tested the significance of treatment effects using likelihood ratio tests comparing nested models. The estimated allometric coefficient was $b = 0.990$ (95% CI: 0.986, 0.994), indicating near-isometric growth. For visualization, we calculated size-corrected growth using the *b*-value estimated from the full model: Scaled Growth = $\log(\text{Final Mass} / \text{Initial Mass}^b)$. We also calculated absolute growth rate as mass change per unit surface area per day: $\text{Growth Rate (mg cm}^{-2} \text{ day}^{-1}) = (\text{Final Mass} - \text{Initial Mass}) / (\text{Total Surface Area} - \text{Wound Area}) / 21 \text{ days}$. This metric represents the mass of calcium carbonate deposited per cm² of coral tissue per day.

To evaluate variation in photosynthetic efficiency (Fv/Fm), we used linear mixed-effects models with the same fixed and random effects structure. For all models we assumed normal error distributions and

homoscedastic variances and evaluated those assumptions using diagnostic residual plots. To assess the significance of fixed effects and their interactions, we used likelihood ratio tests via the `anova()` function from the `car` package. We report chi-squared statistics and associated p-values for all terms.

RESULTS AND DISCUSSION

The effect of coral-associated damselfishes on wound healing depended on wound size (fish $\times$ wound interaction: $\chi^2 = 4.416$, $p = 0.036$, Figure 2). Fish presence increased healing rates for both wound sizes, but the effect was more pronounced for large wounds. For small wounds, fish increased healing rates by 60%, from 0.048 cm²/day (95% CI: 0.027–0.069) without fish to 0.079 cm²/day (95% CI: 0.065–0.092) with fish. For large wounds, fish increased healing rates by 64%, from 0.131 cm²/day (95% CI: 0.094–0.168) without fish to 0.204 cm²/day (95% CI: 0.169–0.239) with fish. Although the proportional increase was similar, the absolute effect of fish was 2.4 times greater for large wounds (0.073 cm²/day) than for small wounds (0.030 cm²/day). Additionally, turf algae frequently colonized wounds during the healing process. By day 21, algae persisted on 8 corals and was biased towards tanks without fish (7/8) and with large wounds (6/8).

The effect of damselfish presence on coral photosynthetic efficiency (Fv/Fm) depended on wound type (fish $\times$ wound interaction: $\chi^2_2 = 10.898$, $p = 0.0043$, Figure 3A). In the presence of damselfish, photosynthetic efficiency was similar across all three wounding treatments: No Wound = 0.686 (95% CI: 0.679–0.693), Small = 0.674 (95% CI: 0.657–0.691), and Large = 0.684 (95% CI: 0.677–0.692). In the absence of fish, however, photosynthetic efficiency declined with increasing wound size: No Wound = 0.684 (95% CI: 0.676–0.691), Small = 0.657 (95% CI: 0.629–0.685), and Large = 0.633 (95% CI: 0.610–0.656). Fish presence had minimal effect on unwounded corals (+0.002 Fv/Fm units) but progressively greater effects on wounded corals: +0.017 for small wounds and +0.051 for large wounds. Thus, damselfish presence positively influenced photosynthetic efficiency, and these effects were most pronounced when wounds were larger and metabolic demands associated with wound repair were presumably higher.

In contrast to effects on healing rate and photosynthetic efficiency, and contrary to previous studies^(9, 2), we observed no significant effects of fish presence, wound size, or their interaction on coral growth after controlling for initial coral size (Fish: $\chi^2 = 0.001$, $p = 0.978$; Wound: $\chi^2 = 1.783$, $p = 0.410$; Fish $\times$ Wound: $\chi^2 = 5.224$, $p = 0.073$; Figure 3B). Given prior studies demonstrating positive effects of fish on coral growth, and our finding that fish increased photosynthetic efficiency and healing rate, we suggest that the absence of detectable calcification responses may reflect the short experimental duration of our study. It may also reflect insufficient sensitivity of the calcification metric or the lack of an effect of acute wounding stress on calcification, but this seems unlikely given past studies^(2, 9, 27).

These findings highlight the critical contribution of coral-associated fishes to coral resilience, specifically their roles in tissue regeneration and physiological recovery following acute injuries. Although the exact mechanisms remain to be elucidated, nutrient subsidies from damselfish excretion likely enhanced symbiotic algal productivity, providing the additional energy required for wound repair^(2, 9). Additionally, turf algae persistence in wounds was markedly more common in corals without fish, suggesting that fish may also facilitate wound repair by mitigating the energetic constraints of competitive exclusion. Consistent with this interpretation, corals without fish also exhibited depressed photosynthetic efficiency; however, this may reflect impaired symbiont performance or reduced tissue availability for symbiont colonization, rather than a direct effect on symbiont physiology alone.

In addition to nutrient benefits, damselfish behavior, such as constant swimming within coral branches, may improve water flow and oxygenation; factors that are particularly important during the nighttime period when metabolic demands of wound healing are high and hypoxia risk increases^(7, 22, 23). Together,

the nutrient and behavioral benefits of coral-associated fishes likely enhance the coral's capacity to regenerate tissue efficiently.

Our results contribute to the emerging perspective that coral-associated fishes significantly enhance coral holobiont resilience, by adding the facilitation of tissue regeneration to the previously established roles of fishes in promoting coral growth and increasing thermal tolerance^(21, 28). By demonstrating that damselfishes actively facilitate coral regeneration following injury, we underscore the functional importance of external symbiotic relationships in buffering corals from both chronic stressors and acute physical disturbances. Preserving these ecological partnerships will likely be crucial for sustaining coral regenerative capacity and enhancing long-term reef resilience.

ETHICS

All procedures involving coral fragments and damselfish (*Dascyllus flavicaudus*) were conducted following institutional and national guidelines for the care and use of animals in research. The study protocol received ethical approval from the University of California, Santa Barbara Institutional Animal Care and Use Committee (Protocol No. 926.1). Research was completed under permits issued by the Government of French Polynesia (Délégation à la Recherche) and the Haut-Commissariat de la République en Polynésie Française (DTRT). We followed best practices to minimize stress and injury to both corals and fishes throughout the experiments.

DATA AND CODE ACCESSIBILITY

Data and analysis code supporting Vega et al.(2025) are available in the Damselfish–Coral Healing Project repository: https://github.com/haydenvega/damselfish_coral_healing_public (v1.0). Cleaned datasets and analysis scripts are licensed under CC BY 4.0. Raw field data are archived within the repository and will be deposited in a long-term archive (e.g., Harvard Dataverse) upon acceptance; DOI will be added to this record when available.

DECLARATION OF AI USE

We have not used AI-assisted technologies in creating this article.

AUTHORS' CONTRIBUTIONS

A.C.S., H.V., and N.M. designed the experiment. H.V. and N.M. conducted the fieldwork and data collection. H.V. analyzed the data with input from A.C.S. and C.W.O. H.V. led the writing of the manuscript. A.C.S., C.W.O. and A.W.S. provided critical edits and helped refine the manuscript. All authors contributed to interpreting the results and approved the final version.

CONFLICT OF INTEREST DECLARATION

We declare no competing interests.

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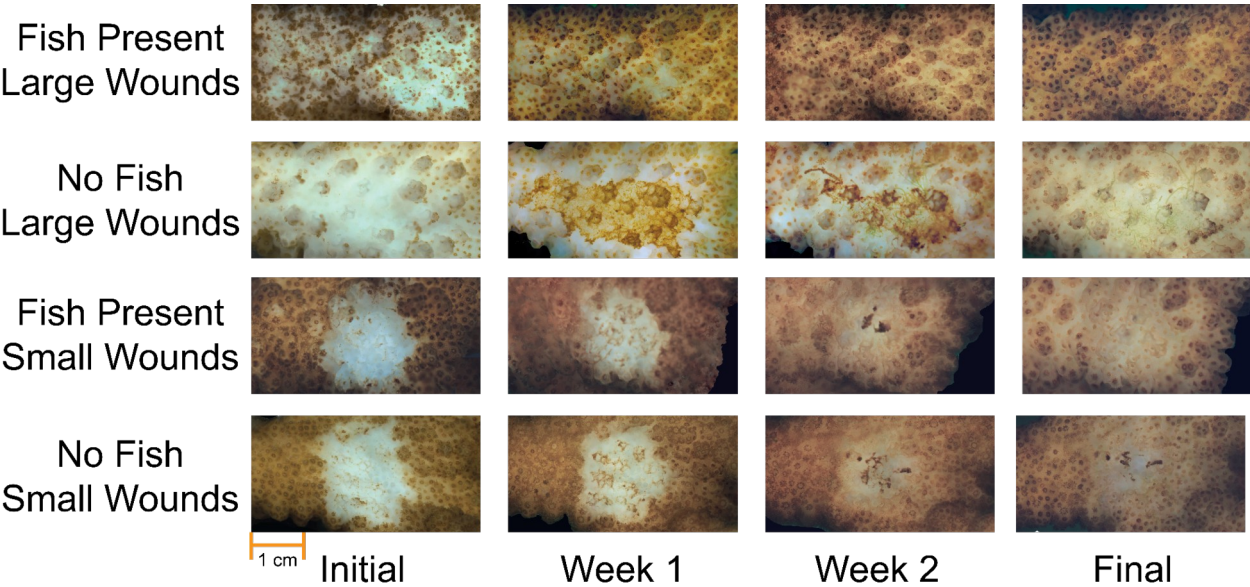
332 FIGURES

333 **Figure 1.** A time series of wound healing by treatment. One coral from each experimental treatment is
334 pictured at four timepoints: initial wounding and one, two, and three weeks post wounding, when the
335 experiment ended. Corals held without fish exhibited less tissue regeneration compared to corals with
336 fish. At three weeks, wounded corals without fish remained only partially healed, while corals with fish
337 exhibited near to or complete healing; the small wound with fish is 98% healed, and the large wound with
338 fish is fully healed.

339 **Figure 2.** Damselfish presence accelerated coral wound healing, particularly for large wounds. Wound
340 healing rate (cm^2/day) in fragments of *Pocillopora* as a function of wound size and fish presence. Small
341 circles represent individual coral fragments ($n = 8$ per treatment); large circles indicate treatment means
342 with vertical bars showing 95% confidence intervals. Corals with resident damselfish (blue, filled circles)
343 exhibited higher healing rates than fish-absent controls (orange, open circles) across both wound size
344 treatments. This effect was most pronounced in corals with large wounds, where corals with fish healed at
345 approximately twice the rate of corals without fish.

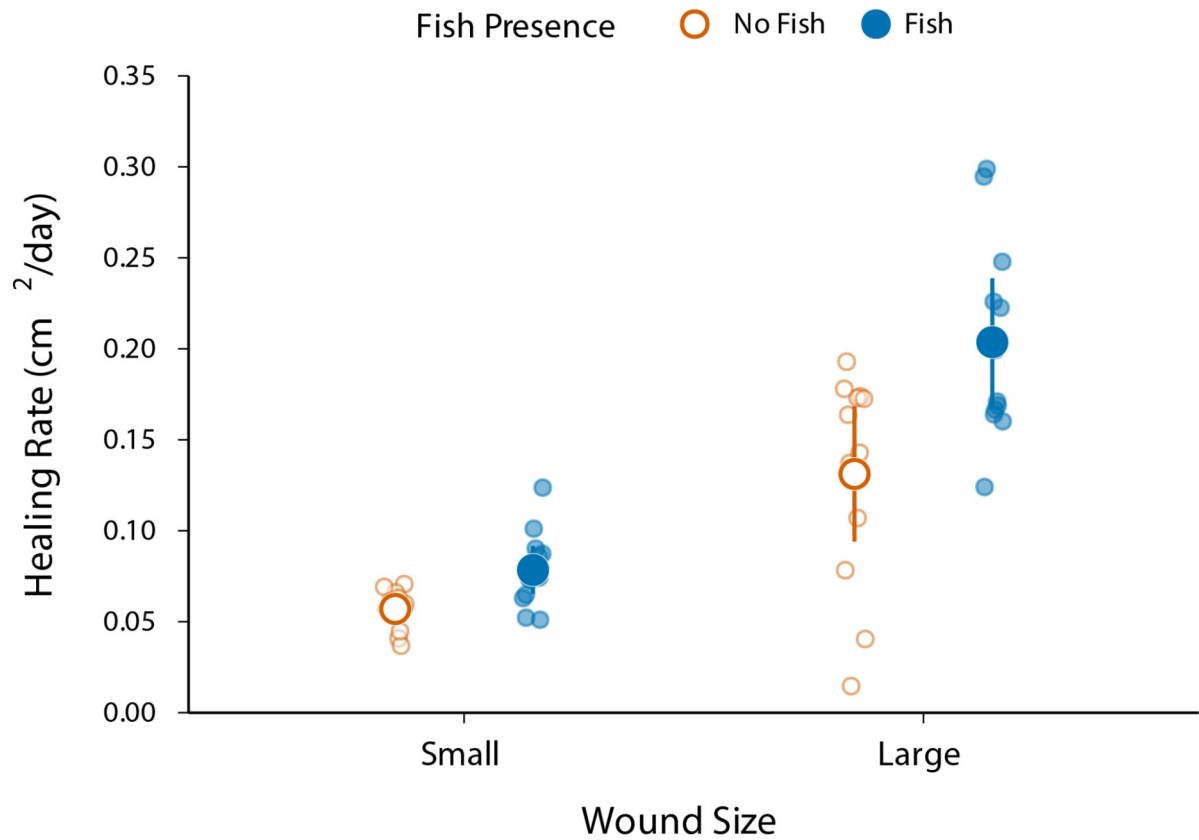
346 **Figure 3.** Effects of damselfish on coral growth rate and photosynthetic efficiency across wound
347 treatments. (A) Coral skeletal growth rate ($\text{mg day}^{-1}\text{cm}^{-2}$) and (B) maximum quantum yield of
348 photosystem II (F_v/F_m) measured in fragments of *Pocillopora* exposed to three wound size treatments
349 (No Wound, Small, Large) in the presence (blue, filled circles) or absence (orange, open circles) of
350 *Dascyllus* damselfish. Small circles represent individual coral fragments; large circles indicate treatment
351 means $\pm$ 95% confidence intervals. (A) Growth rates were similar across all treatments. (B) Fish presence
352 had minimal effect on photosynthetic efficiency in unwounded corals, but enhanced F_v/F_m in wounded
353 corals, with the largest difference observed in corals with large wounds. Corals without fish showed
354 reduced F_v/F_m in wounded treatments, particularly in the large wound treatment where corals without
355 fish exhibited the lowest photosynthetic efficiency.

356 **Figure 1.**



357

358 **Figure 2.**



359

360

361 **Figure 3.**

